

SageHaLow

A Low-Cost Modular Environmental Sensing Platform Using Wi-Fi HaLow and the SAGE Edge Infrastructure

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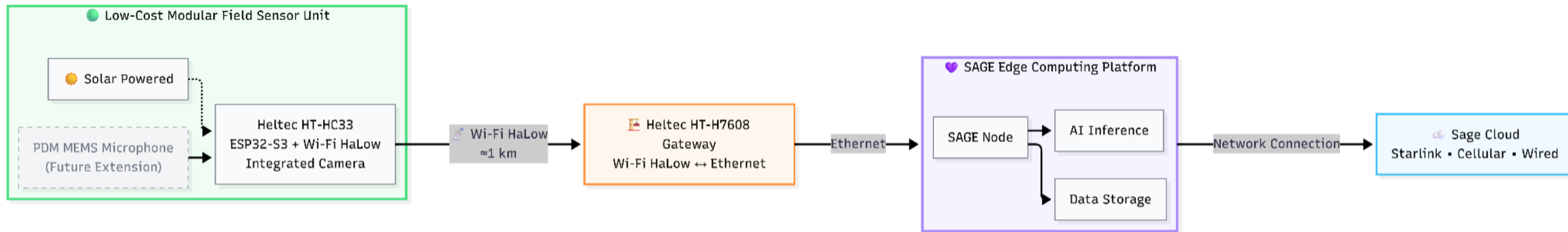
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A proof-of-concept demonstrating long-range Wi-Fi HaLow communication between low-cost sensing devices and the SAGE Edge Platform.



System Architecture

- Low-cost modular environmental sensing platform
- Long-range Wi-Fi HaLow communication (~1 km)
- Integration with the SAGE Edge Platform



Project Accomplishments

Milestone	Outcome
Hardware	Configured and tested the Heltec HT-HC33 camera platform.
Communication	Established long-range Wi-Fi HaLow communication through the HT-H7608 gateway.
SAGE Integration	Transferred images to the SAGE Edge Platform.
System Validation	Demonstrated a low-cost environmental sensing architecture.

Future Development

Near-Term Development

- PDM MEMS microphone integration
- Event-driven image capture

Long-Term Vision

- Edge AI Inference
- Scalable environmental monitoring

The modular SageHaLow architecture provides a scalable foundation for future AI-enabled environmental monitoring.

Learn More

GitHub Repository

github.com/hairikhs/sage-summer-camp-2026

Repository Contents

- Project Report
- System Architecture
- Firmware Documentation
- Future Development

Project materials and future updates are available in the GitHub repository.